



COMP 003 – COMPUTER PROGRAMMING 2
SECOND SEMESTER, AY 2025-2026

FINAL LABORATORY PROJECT

Name: _____	Score: _____
Name: _____	
Name: _____	Program: _____
Name: _____	Course Code - Section: _____
Name: _____	Faculty <u>Asst. Prof. Ma. Ymelda Batalla</u>

Final Laboratory Project Title: **Menu-Driven Retail Inventory Management and Customer Ordering System Using C++ Structures and File Handling**

- Learning Outcomes:**
- Upon completion of the project, the students should be able to:
- **Apply** the fundamental concepts of structures, functions, conditional statements, loops, and file handling in C++ **programming**.
 - **Analyze** program behavior through systematic testing, debugging, and validation to ensure accurate inventory processing and data integrity.
 - **Design and develop** a complete menu-driven inventory management system that integrates inventory control, customer order processing, permanent data storage, and established programming standards and documentation practices.
 - **Evaluate and interpret** program outputs and results to verify accuracy, reliability, and consistency with system requirements.
 - **Collaborate effectively** with team members to plan, develop, and accomplish the assigned group project.

- Instructions:**
- This final project is composed of five (5) students per group.

Project Scenario

A small store company is experiencing difficulties in manually tracking product inventories and processing customer orders. The store owner would like to have a computerized solution that can efficiently manage product information, monitor stock availability, process customer purchases, and maintain accurate inventory records.

As a software developer team, your task is to design and develop a **menu-driven Inventory Management and Customer Ordering System** using **C++**, applying the concepts of **conditional statements, loops, functions, data structures, file handling, and input validation and error handling**. The system should allow users to add, update, view, search, and delete product records; monitor current stock levels; accept customer orders; automatically update inventory quantities after each successful transaction; and store inventory data permanently save in a file for future use.

The completed application should demonstrate proper programming standards, modular design, data validation, and efficient inventory processing while providing an accurate and user-friendly solution for the store's daily operations.

- Functional Requirements**
- The system must be able to:
 - Provide a menu-driven interface for user interaction.
 - Add new products to the inventory.
 - Update existing product information.
 - Delete product records from the inventory.
 - Search for a specific product.
 - Display all available products and their details.
 - Monitor and display current stock quantities.
 - Accept and process customer orders.
 - Validate stock availability before confirming an order.
 - Automatically deduct inventory quantities after a successful order.
 - Save and retrieve inventory records using file handling.
 - Generate transaction summaries or order receipts.

- Final Project Documentation Format**
- Cover Page
 - Table of Contents
 - Introduction
 - Project Objectives
 - Program Design
 - Data Structure Design



- Data/File Repository Design
- Menu Structure
- Process Flow
- System Features and Source Code
- Sample Output
- Challenges encountered and solutions
- Conclusion

Final Project Evaluation Rubric

Criteria	Rating (1-5)	Weighted Score
Functional Requirements (x10) - All required features are fully implemented, properly integrated, and function correctly as a complete system.		
User Interface (3) - The menu system is well-organized, easy to navigate, and clearly presented to ensure good readability and user experience.		
Validation and Error Handling (x3) - Appropriate input validation is applied, with proper error handling to ensure system reliability and prevent invalid inputs.		
Documentation (x2) - Documentation is complete, well-organized, and properly formatted, with clear presentation of all required project component.		
Project Presentation and Demonstration (x1) - Demonstrates clear understanding of the system by effectively explaining its design, demonstrating all functionalities, and answering questions confidently.		
Teamwork and Contribution (x1) - peer assessment and documented evidence of active participation and contribution from all group members.		
TOTAL		/ 100